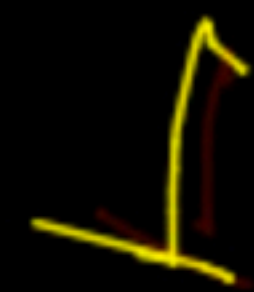


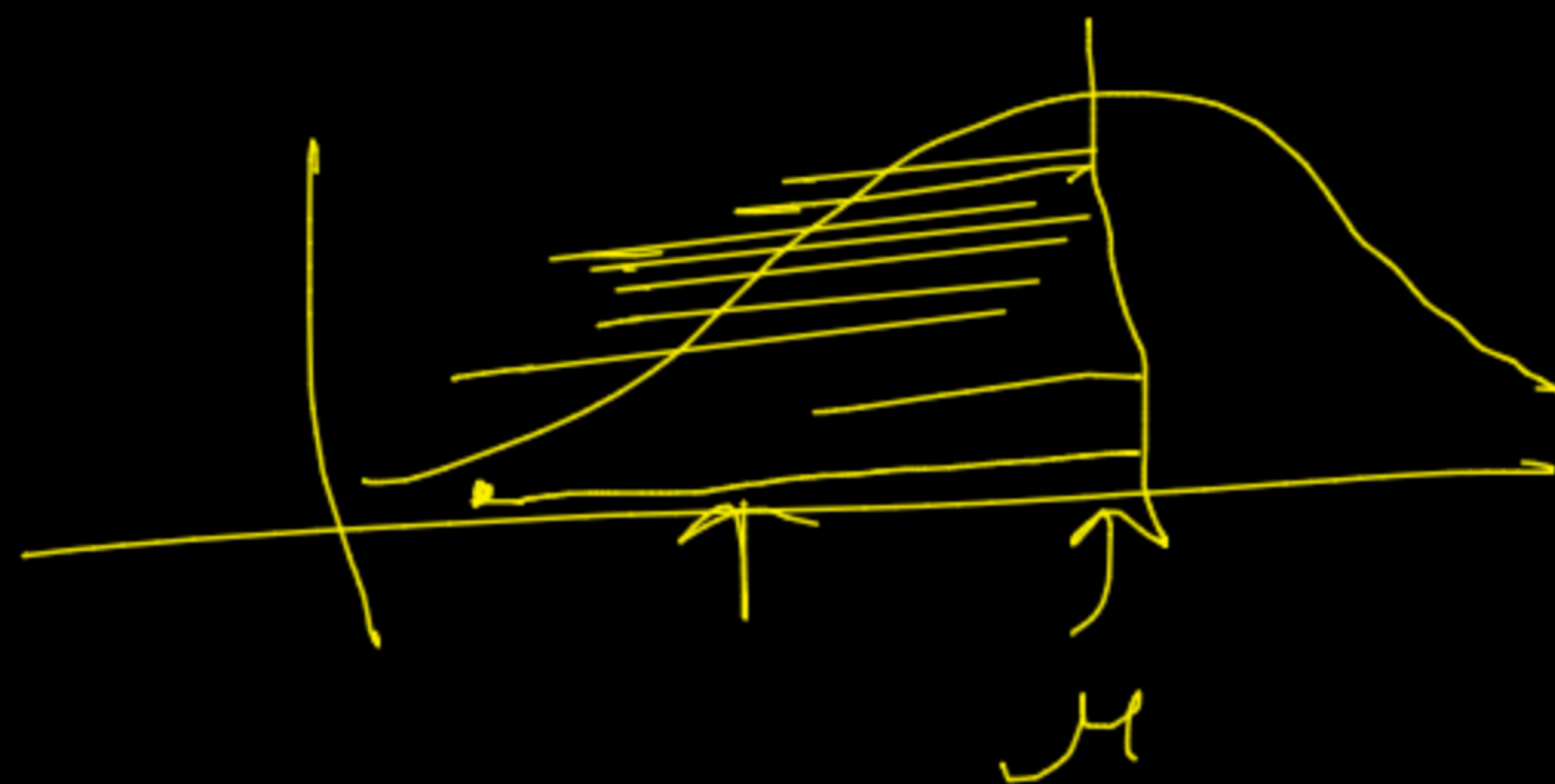
17 Negative skewness

Left skewed



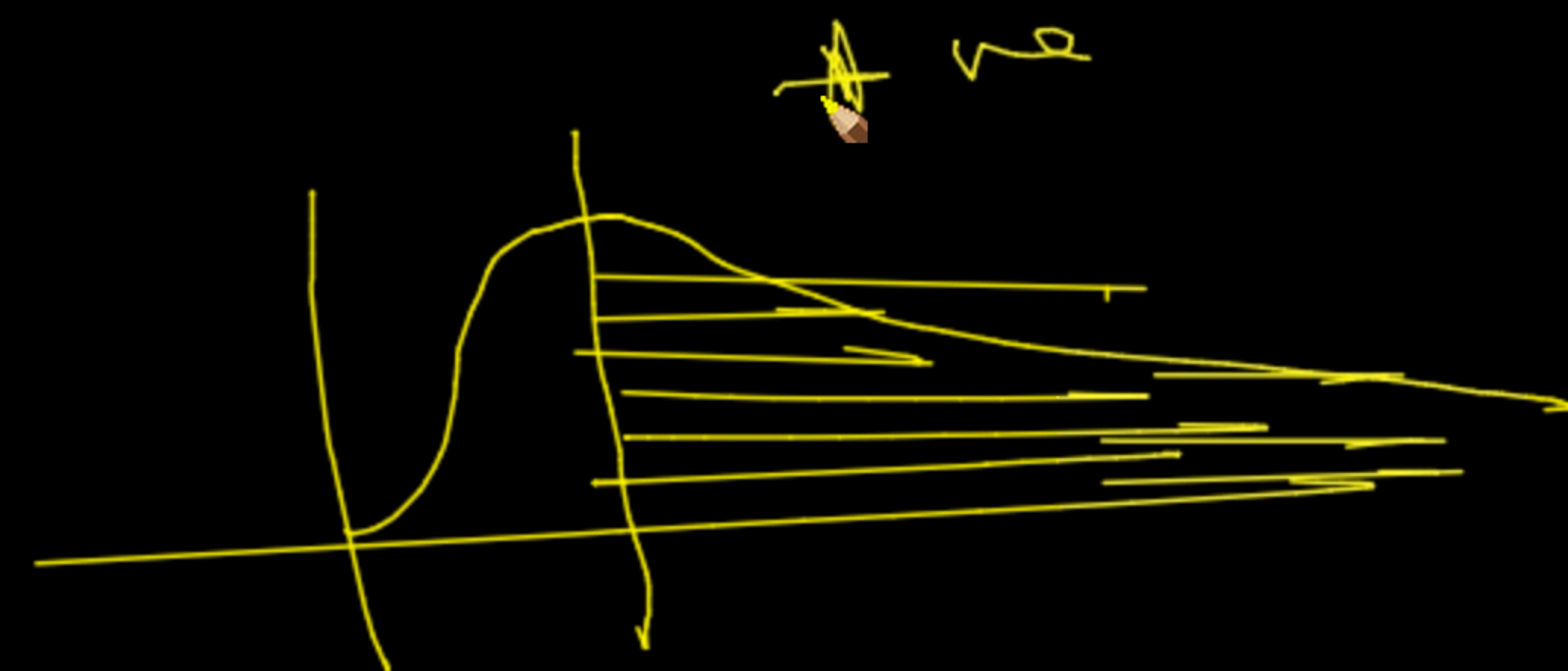
$\mu \rightarrow$ location where central value lies

$\sigma \rightarrow$ spread



17 Positive skewness

Right skewed



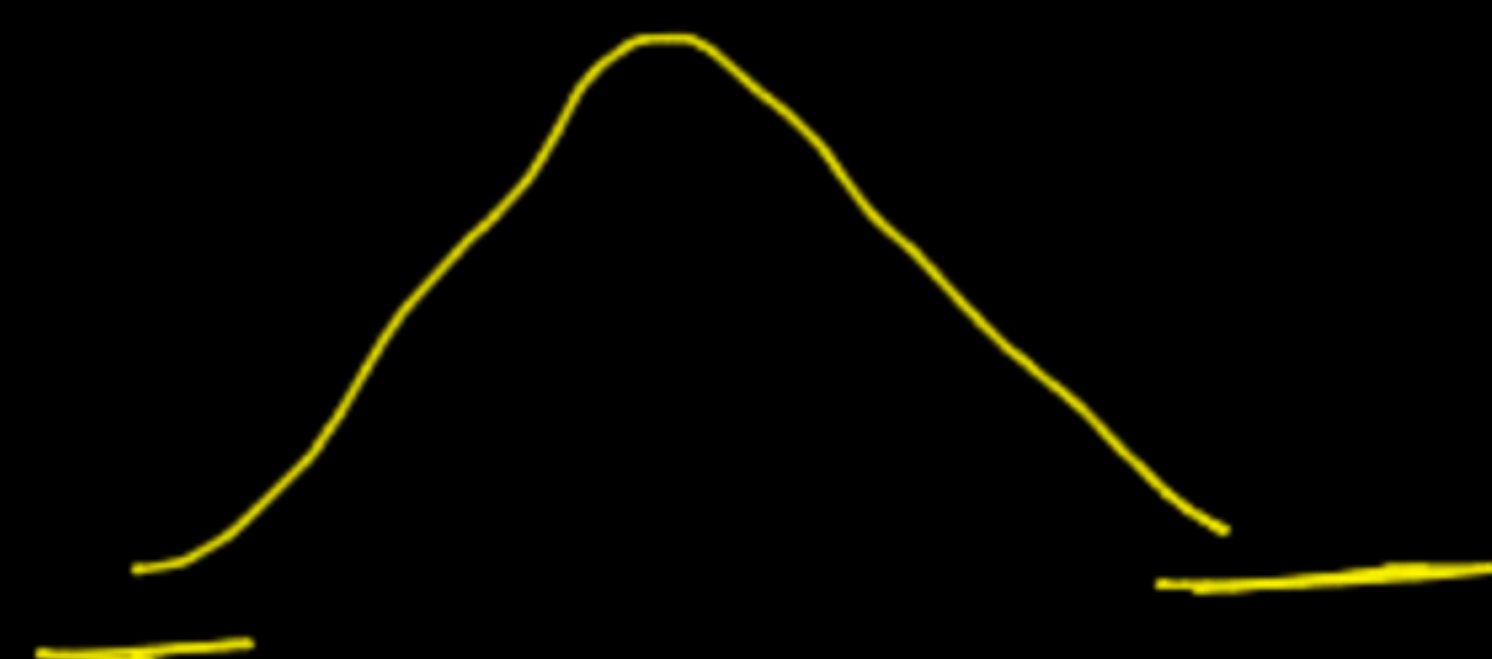
Kurtosis



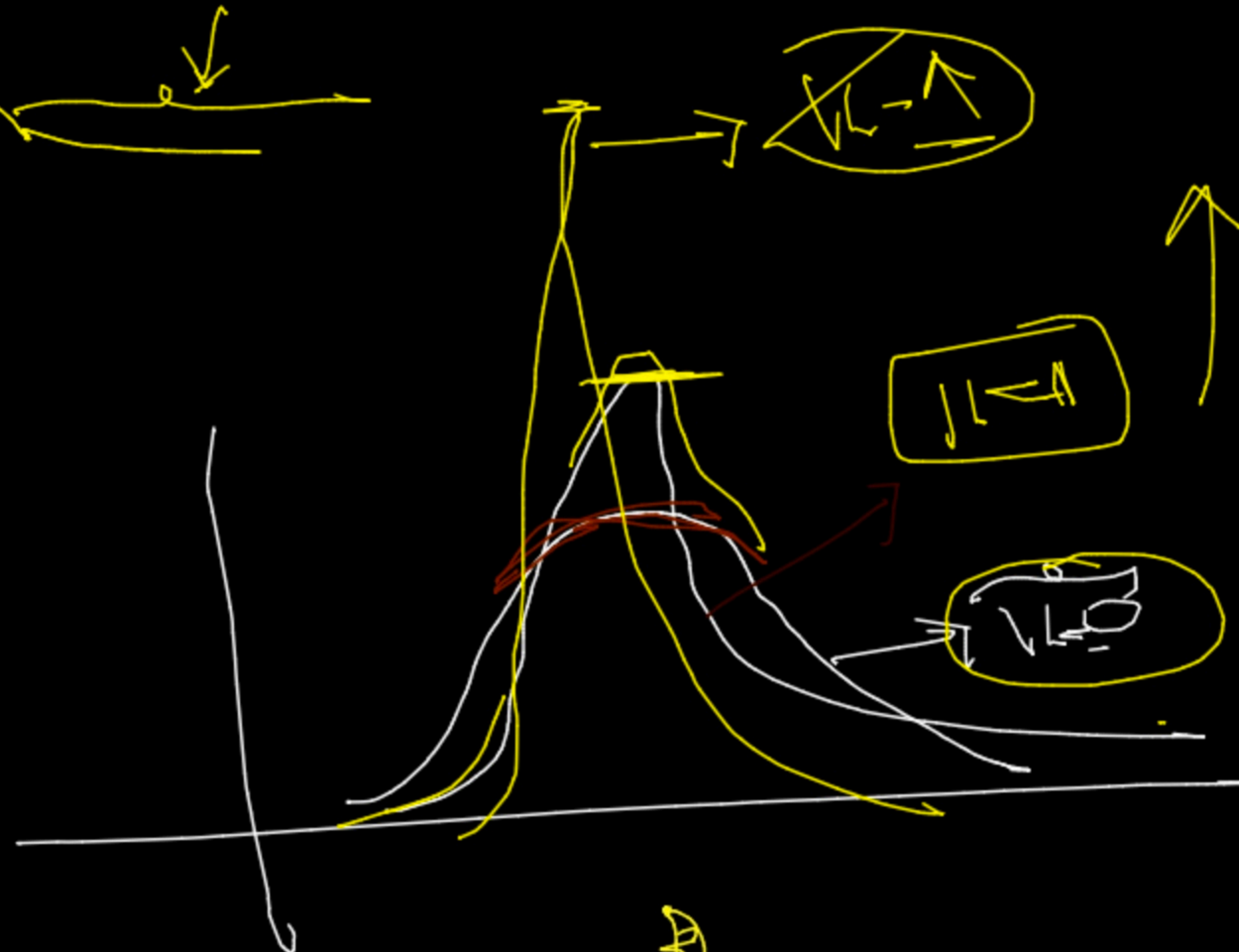
Normal (G.D.)

$\sigma \uparrow$

σ



→ Peakness (Sharpness)



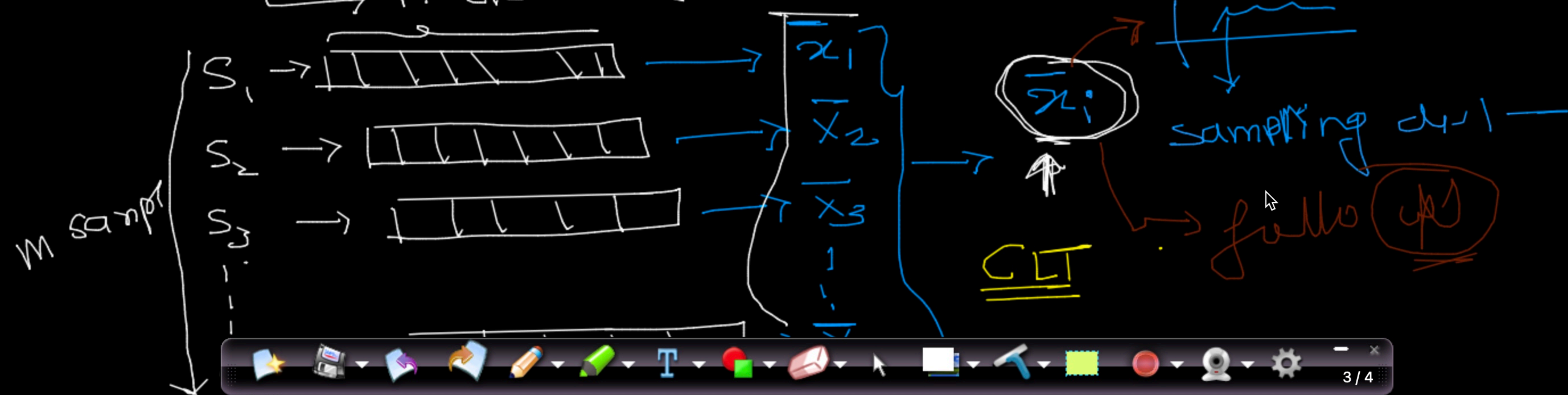
Sampling distribution



→ m Random sample

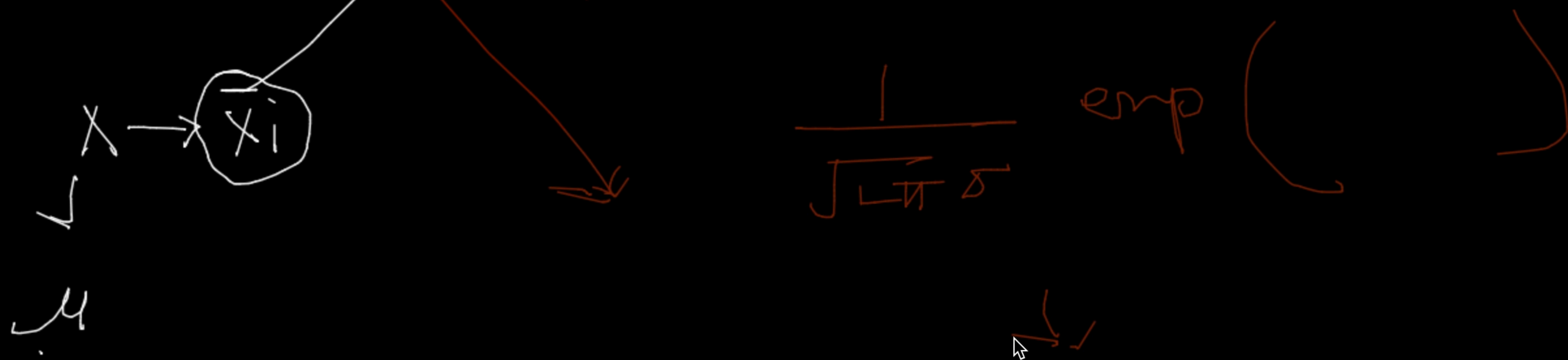
n element (30)

'sample mean'



If X (original population) has finite μ and σ^2 then

$\bar{X}_n \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$ as $n \rightarrow \infty$ $X \sim N(\mu, \sigma)$

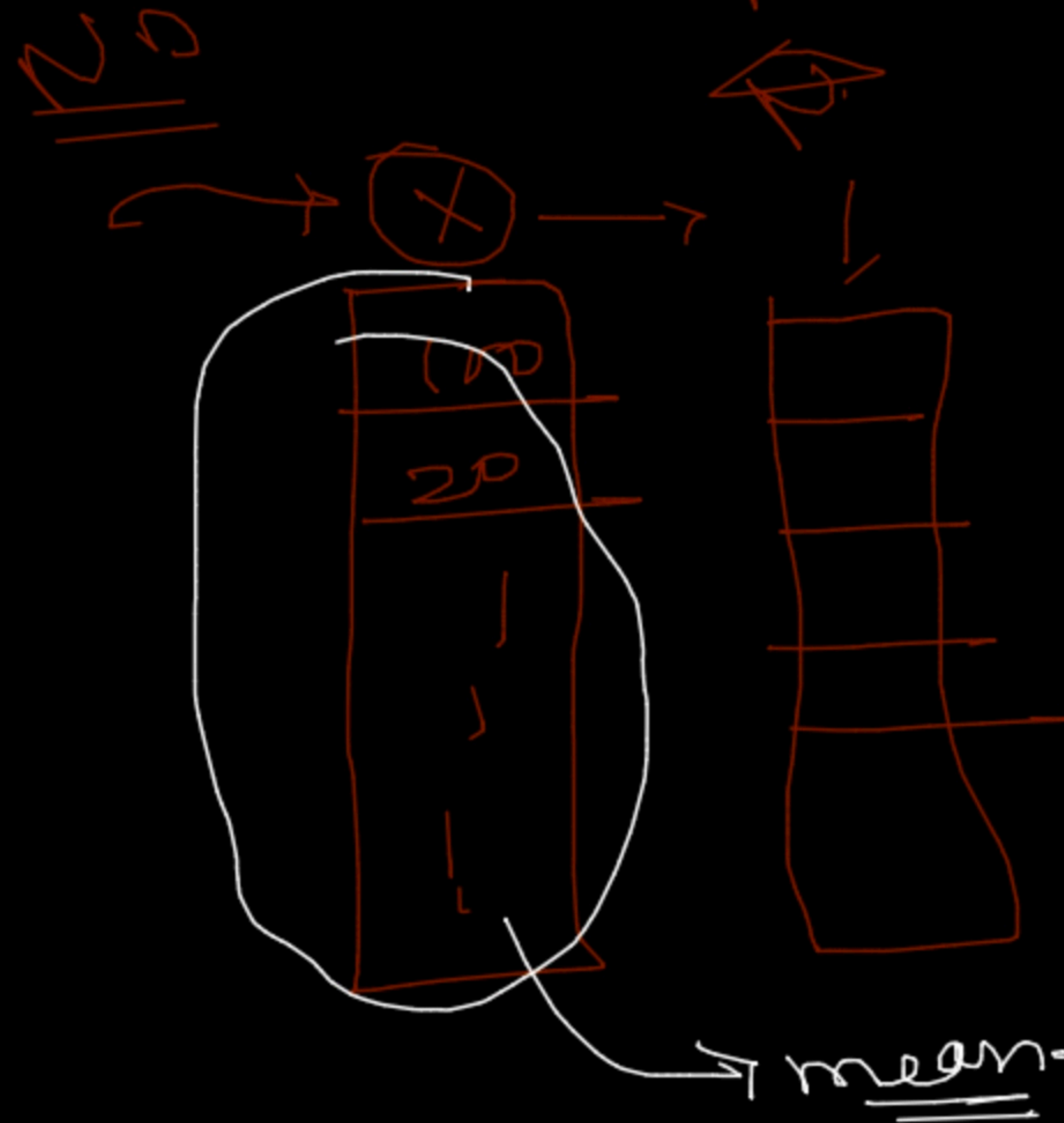


Standard Normal variable

$$Z \sim (0, 1)$$

$$X \sim N(\mu, \sigma)$$

$$x_i' = \frac{x_i - \mu}{\sigma}$$



$$\frac{100 - 120}{15} \rightarrow$$

$$120 \quad \sigma = 15$$